

Rishabh Agarwal

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EDUCATION

Columbia University

M.S. in Computer Science, Machine Learning Track, GPA: 4.00/4

New York City, NY

May 2026 – Expected May 2028

Coursework: Discrete Math & Probability, Data Structures & Algorithms, Linear Algebra

University of California, Los Angeles

B.S. in Neuroscience, GPA: 3.91/4

Los Angeles, CA

Aug. 2021 – Aug. 2024

Coursework: Multivariable Calculus, Statistics, Python Programming

TECHNICAL SKILLS

Languages: Python, C, C++, Swift, Java, JavaScript, R, SQL, HTML5, LaTeX, Z Shell

Frontend + Backend: Next.js, REST APIs, React.js, Node.js, SwiftUI, TypeScript, Tailwind CSS, Flask

Tools: Git, Docker, Unix, Linux, Claude Code, XCode, VS Code, Anaconda, Pytest

Data & Machine Learning: Pandas, NumPy, PyTorch, ETL Pipelines, Data Modeling & Validation

Concepts: Object-Oriented Design, IOS & Full-Stack Web Development, Distributed Data Processing, TCP/IP

EXPERIENCES

Columbia AI Alignment (CAIAC)

New York City, NY

Technical Fellow, AI Safety

Sep. 2026 - Current

- Analyzing research on **mechanistic interpretability**, **RLHF** & reward hacking, **goal misgeneralization**, **AI control**, **scalable oversight**, & **red teaming** w/ Columbia faculty & researchers in a **semester-long fellowship**

SH Allergy, Asthma & Immunology

San Francisco, CA

Medical Assistant

Sep. 2025 - May 2026

- Shipped **end-to-end fullstack web app** managing **100+** staff assignments and holiday requests across **5 clinics**, reducing **~80%** of staffing errors and shortening weekly schedule revisions to **instant chat/email updates**
- Designed **RESTful backend** in **Flask (Python)** serving a **React (TypeScript)** web app; integrated **Resend API** and **Google Chat Webhooks** for automated email and chat notifications to staff
- Gspread & Pandas** migrate and sync the legacy manual-spreadsheet workflow into the application's data layer

UCLA Health Brain-Gut Interaction Lab - Dr. Arpana Church

Los Angeles, CA

Data Analyst/Researcher

Aug. 2022 – July 2024

- Enabled **4** downstream research projects & authored Nature publication by building a **Python** ETL pipeline for the 12K-subject National Institutes of Health (NIH) Adolescent Brain Cognitive Development (ABCD) Study dataset
- Utilized **Selenium & BeautifulSoup** for **web-scraping**/extraction of micro-biome data, and **Pandas & Numpy** packages for data cleansing, merging ABCD and non-ABCD data, and data standardization across 4 studies
- Doubled** center's output and migrated **4 projects** off legacy SPSS software by engineering an **R framework** for portable **structural equation modeling (SEM)**, using **Lavaan**, **Matplotlib**, **Tidyverse** packages
- Led Neuroscience capstone from concept to manuscript while **mentoring three junior researchers** in statistical analysis, presenting at conferences like **Annals of Neurology**, SCORE symposium, and UCLA Neuro week

PROJECTS

Bench Adoption Webapp | *Next.js, TypeScript, tailwind CSS, PostgreSQL, Node.js*

- Built web-app to view, adopt, and (admin) manage 470+ benches for NYC Van Cortlandt Park's donor program
- Supabase auth and **Postgres** backend with **Next.js** server actions and a role-protected admin dashboard

Chilli Dog Oat Milk Latte (IOS App) | *Swift, SwiftUI, Xcode*

- Built a native iOS multiplayer card game in **Swift** using **SwiftUI** for real-time, touch-driven card interactions
- Fully offline pass-and-play experience, letting groups play anywhere without a physical deck

NIH ABCD Study Data Dictionary | *Python, BeautifulSoup, Selenium, Pandas, NumPy*

- Dashboard to index & navigate the ABCD Study's 90,000 variables and 700,000 MRI scans spanning 100 terabytes
- Made **hierarchical tree data structure** mapping variable names, descriptions, data types, and package sources
- Cut research data-discovery time from hours to minutes by surfacing relevant variables

PUBLICATIONS

- An E, Delgadillo DR, Yang J, **Agarwal R**, Et al., Stress-Resilience Impacts Psychological Wellbeing: Evidence from Brain-Gut Microbiome Interactions. Nat Ment Health (2024). doi: 10.1038/s44220-024-00266-6.